Review notes · for you

Goals and limits: unit check

Optimization · Level 2 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Choices, limits and a goal
- Draw the plans that fit
- Whole numbers only
- Tell the computer

Activity: Model swap

1. Recall (3 min): Write one idea from each of the four lessons, in your own words.
2. Write (7 min): Write a model for a choice you make (snacks, homework time, a team): variables, at least two limits and a goal, plus two plans.
3. Swap (7 min): Swap with a partner. Check their plans against every limit, and add one rule their model is missing, such as a fairness rule.
4. Share (3 min): Share the best rule someone added to your model, and why it matters.

Discussion

- Which idea from this unit surprised you most?
- When is the cheapest or biggest plan not the best one?
- How would you explain a model to a younger student?

Check yourself

- One model I could write this week, and its goal.

Try it this week
Pick one choice you make every week and write it as a model: variables, limits and a goal. Check two plans, and add one rule that makes it fair to someone else.

Quiz sheet

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Name _____ Class _____ Date _____

1 **In a model, what is a variable?**

◯ A rule every plan must keep ◯ The number you make as big as possible ◯ A choice you decide, with a unit

2 **The garden: x tomato beds, y bean beds. Match each rule to its maths.**

Land: at most 8 beds	• • • •	2x + y ≤ 14
Water: at most 14 cans a day	• • • •	30x + 40y
The profit to make as big as possible	• • • •	x + y ≤ 8

3 **The plan 6 tomato beds and 2 bean beds keeps all three garden limits (land 8, water 14, hours 18).**

◯ True ◯ False

4 **In a linear program, where is the best plan?**

◯ At a corner of the shaded shape ◯ Just outside the shaded shape ◯ In the middle of the shaded shape

5 **Which choices can only be whole numbers or yes-or-no? Choose all that apply.**

☐ How many tomato beds to plant ☐ How many buses to book ☐ Litres of juice for the party

☐ Whether the desks go on the truck

6 **Complete the garden model: the land limit and the goal.**

```
model. _____(x + y _____ 8)
model.add(2 * x + y <= 14)
model.add(x + 3 * y <= 18)
model. _____(30 * x + 40 * y)
```

Word bank: `minimize` `add` `<=` `print` `maximize` `>=`

7 **Put the steps of solving the garden on graph paper in order.**

☐ Mark the corners

☐ Draw each limit as a line

☐ Shade where every limit is kept

☐ Pick the best corner

☐ Work out the profit at each corner

8 **Two cups of rice give 260 kcal each. What does this program print?**

```
rice = 2
energy = 260 * rice
print(energy >= 700)
```

◯ False ◯ True ◯ 520

9 **Complete the sentence about the truck.**

The quick rule took the _____ panels first and scored 60, but desks and _____ score 65: a _____ rule is not always best.

Word bank: books · greedy · corner · solar · rounding

10 **If the solver prints INFEASIBLE, running it again will find a plan.**

◯ True ◯ False

Check your answers

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1 **A choice you decide, with a unit**

A variable is a choice, like r = cups of rice.

2 **Land: at most 8 beds — $x + y \leq 8$**
Water: at most 14 cans a day — $2x + y \leq 14$
The profit to make as big as possible — $30x + 40y$

Land counts beds of both kinds; water counts 2 cans for each tomato bed and 1 for each bean bed; the profit has no sign because it is the goal, not a limit.

3 **True**

Land 6 + 2 = 8, water 12 + 2 = 14 and hours 6 + 6 = 12: all kept.

4 **At a corner of the shaded shape**

Slide the profit line up and the last point of the shape it touches is a corner.

5 **How many buses to book; Whether the desks go on the truck; How many tomato beds to plant**

Buses and garden beds come whole, and a load is taken or not.

6

```
model.add(x + y <= 8)
model.add(2 * x + y <= 14)
model.add(x + 3 * y <= 18)
model.maximize(30 * x + 40 * y)
```

A limit is added with model.add, "at most 8" is <=, and the most profit is maximize.

7 **Top to bottom: 3, 1, 2, 5, 4**

The lines make the shaded shape, the shape has corners, and the best plan is the corner with the most profit.

8 **False**

260 × 2 = 520, and 520 is not at least 700, so the comparison prints False: two cups of rice alone don't meet the energy limit.

9 **The quick rule took the solar panels first and scored 60, but desks and books score 65: a greedy rule is not always best.**

The greedy rule grabbed the solar panels (the most points per kilo), which left room only for small loads.

10 **False**

INFEASIBLE is a proof that no plan keeps every rule.